SIMATS ENGINEERING

CSA0705 - COMPUTER NETWORKS

STUDENT NAME : BALAJI A

REGISTER NO : 1911260245

DEPARTMENT : B.TECH COMPUTER SCIENCE ENGINEERING

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FACULTY NAME : KUMARAN G

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1. Introduction

This assignment analyses the network topology described in the contractor's drawing. The original design consists of three hubs (each connecting eight PCs), a 24-port main switch, a router with interfaces G0/0 and G0/1, a second switch, and twelve IP phones. The objectives are to determine the number of collision domains and broadcast domains, illustrate the topology with domain markings, explain the OSI layers of the devices, discuss the behaviour of hubs, identify connection types, present relevant standards, and evaluate the benefit of replacing hubs with unmanaged switches.

2. Original Network – Collision and Broadcast Domains

2.1 Network Description

The contractor's network is structured as follows:

- Three hubs, each connected to 8 PCs (24 PCs in total).
- All three hubs connect to one 24-port switch (Main Switch).
- The Main Switch connects to Router interface G0/0.
- Router interface G0/1 connects to a second switch (Switch-2).
- Switch-2 connects to 12 IP phones.

2.2 Calculation of Collision Domains

A collision domain is a network segment in which data packets can collide with one another. Key principles applied:

- A hub operates at Layer 1 and creates a single shared collision domain for all connected devices.
- Each port on a switch is a separate collision domain.
- Each router interface forms its own collision domain on the connected segment.

Applying these rules to the original design yields:

Segment / Connection	Collision Domains	Reason
Three hub segments (Hub-1, Hub-2, Hub-3)	**3**	One shared CD per hub
Main Switch ↔ Router G0/0	**1**	Point-to-point link
Router G0/1 ↔ Switch-2	**1**	Point-to-point link
Twelve IP phone ports on Switch-2	**12**	One CD per switch port
TOTAL	**17**	

Note: The uplinks from the three hubs to the Main Switch are already accounted for within each hub's single collision domain; they do not create additional collision domains because the hub shares the medium.

2.3 Calculation of Broadcast Domains

A broadcast domain is a logical division of a network in which all nodes can reach each other by broadcast at the data-link layer. Routers do not forward Layer-2 broadcasts; therefore each router interface (and the network attached to it) constitutes a separate broadcast domain.

- Broadcast Domain 1: Everything reachable via Router interface G0/0 (Main Switch, three hubs and 24 PCs).

- Broadcast Domain 2: Everything reachable via Router interface G0/1 (Switch-2 and 12 IP phones).

Total broadcast domains = **2** (assuming no VLANs are configured).

3. Original Network Topology Diagram

Figure 1 shows the complete original topology with collision domains and broadcast domains clearly marked. Dashed red outlines indicate collision domains; dotted blue/green outlines indicate the two broadcast domains separated by the router.

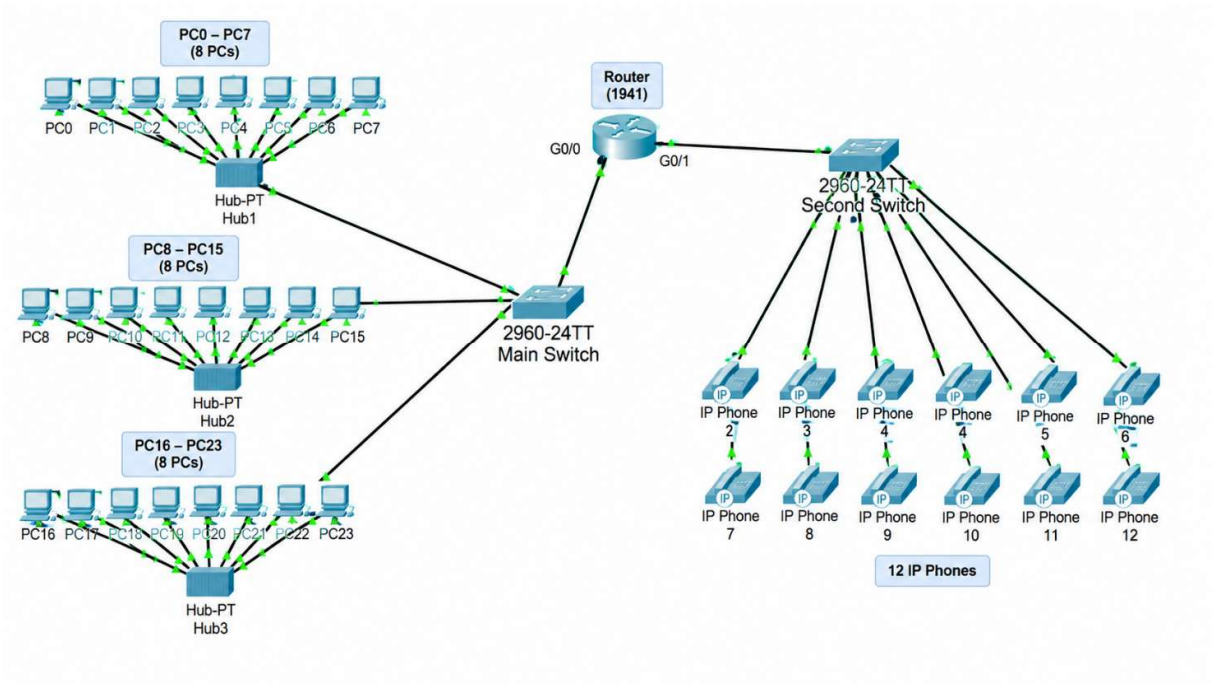


Figure 1: Original Network Topology with Collision and Broadcast Domains

4. OSI Layer of Devices

Each device in the topology operates at one or more layers of the OSI reference model:

Device	OSI Layer(s)	Explanation
Hub	Layer 1 (Physical)	Repeats electrical signals to all ports. No addressing or filtering. Creates a single shared collision domain.
Switch	Layer 2 (Data Link)	Forwards frames based on MAC addresses. Each port is an independent collision domain. Does not separate broadcast domains (unless VLANs are used).
Router	Layer 3 (Network)	Routes packets based on IP addresses. Does not forward Layer-2 broadcasts, thereby separating broadcast domains.
PC	Layers 1–7	End host implementing the full protocol stack (Physical through Application) to run user applications.
IP Phone	Layers 1–7	End device that implements the complete OSI stack to support VoIP signalling and media streams.

5. Why Hubs Cannot Prevent Collisions

A hub is a multiport repeater that operates exclusively at the Physical layer (OSI Layer 1). Its fundamental behaviours explain why collisions cannot be prevented:

1. **Signal regeneration only:** The hub receives a signal on one port and retransmits an identical electrical signal on all other ports. It has no knowledge of MAC addresses or frame structure.
2. **Shared medium:** All devices attached to a hub share the same collision domain. When two stations transmit simultaneously, their signals collide and are corrupted.
3. **No buffering or store-and-forward:** Unlike a switch, a hub cannot hold a frame while waiting for the medium to become free; it simply floods the signal.
4. **CSMA/CD is still required:** Devices must detect collisions and retransmit. The hub itself provides no collision-avoidance mechanism.

Consequently, the bandwidth of the segment is shared among all attached stations, and performance degrades rapidly as the number of active devices increases.

6. Point-to-Point and Multipoint Connections

Connections in the topology can be classified as follows:

Point-to-Point Connections

A point-to-point connection links exactly two devices. In the given network these include:

- Main Switch ↔ Router G0/0
- Router G0/1 ↔ Switch-2
- Each individual PC ↔ its attached hub (physical cable)
- Each IP phone ↔ Switch-2 port

Multipoint (Shared) Connections

A multipoint connection allows more than two devices to share the same communication medium. In the original design the multipoint segments are the three hub-based collision domains: all eight PCs attached to a given hub, together with the hub's uplink to the Main Switch, share a single logical multipoint medium. Although each physical cable is point-to-point, the hub electrically joins them into one collision domain, creating a logical multipoint topology.

On the switched segments (Main Switch and Switch-2), each port is a dedicated point-to-point link; there is no shared multipoint medium at Layer 1/2.

7. Networking Standards

The following table summarises the principal standards relevant to the devices and protocols used in modern enterprise networks:

Technology	Standard / Body	Description
Ethernet	IEEE 802.3	Defines the Physical and Data-Link layers for wired LAN technology, including frame format, CSMA/CD, and successive speed generations (10 Mb/s to 400 Gb/s).
Wi-Fi	IEEE 802.11	Wireless LAN standard covering physical layer modulations, MAC protocols, security (WPA3), and successive amendments (a/b/g/n/ac/ax/be).

IPv4	IETF RFC 791	Internet Protocol version 4 – connectionless packet delivery, 32-bit addressing, fragmentation and reassembly.
OSI Model	ISO/IEC 7498	Seven-layer reference model that standardises the functions of communication systems and guides protocol design.
5G	3GPP specifications & ITU IMT-2020	Fifth-generation mobile system defined by 3GPP Releases 15+ and recognised by ITU under the IMT-2020 framework for enhanced mobile broadband, ultra-reliable low-latency and massive machine-type communications.

8. Why Industries Should Purchase Standards-Compliant Equipment

Purchasing networking equipment that conforms to open standards yields several critical advantages:

1. **Interoperability:** Devices from different vendors can interoperate reliably, avoiding proprietary lock-in and enabling multi-vendor architectures.
2. **Future-proofing:** Standards evolve in a controlled manner; compliant equipment can usually be upgraded or extended without wholesale replacement.
3. **Security and reliability:** Standards bodies incorporate security best practices and rigorous testing requirements that proprietary solutions may lack.
4. **Cost efficiency:** A competitive market of standards-compliant products drives prices down and improves quality.
5. **Regulatory and contractual compliance:** Many industries and government contracts mandate the use of standardised, certified equipment.

9. Alternative Design – Hubs Replaced by Unmanaged Switches

9.1 Topology Change

In the alternative design the three hubs are replaced by three unmanaged Layer-2 switches (Access SW-1, Access SW-2 and Access SW-3). All other devices and connections remain unchanged. Figure 2 illustrates the revised topology.

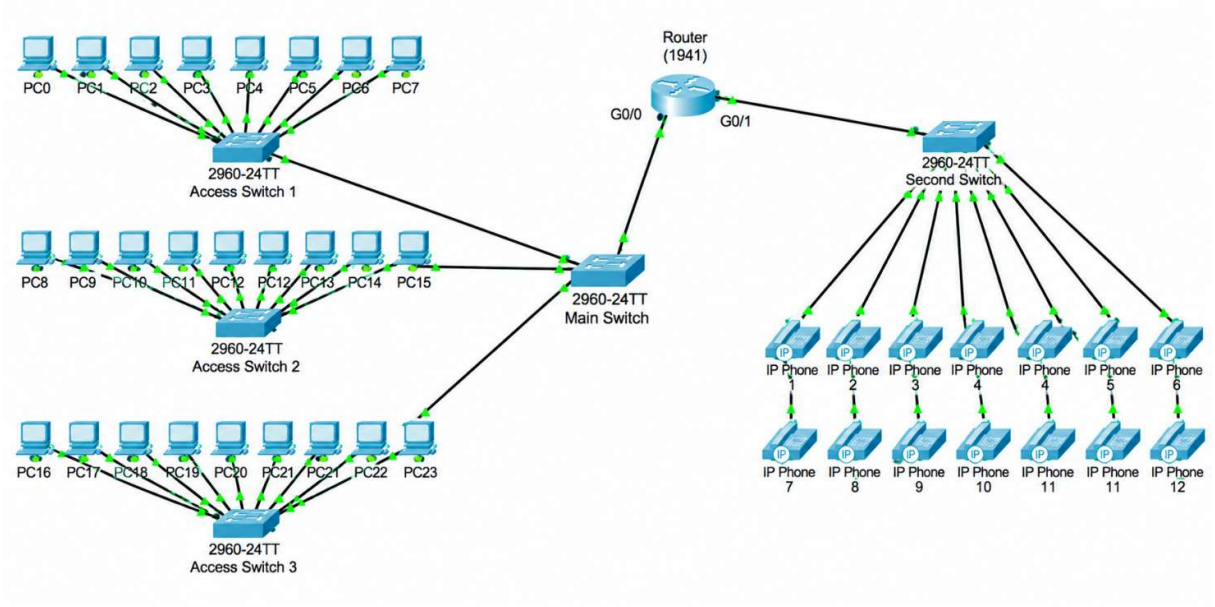


Figure 2: Alternative Network Topology (Hubs replaced by Unmanaged Switches)

9.2 Recalculation of Domains

With unmanaged switches, every switch port becomes an independent collision domain:

Segment / Connection	Collision Domains	Reason
24 PC-to-access-switch ports	**24**	One CD per switch port
Three access-switch uplinks to Main Switch	**3**	One CD per uplink
Main Switch ↔ Router G0/0	**1**	Point-to-point link
Router G0/1 ↔ Switch-2	**1**	Point-to-point link
Twelve IP phone ports on Switch-2	**12**	One CD per switch port
TOTAL	**41**	

Broadcast domains remain **2**. Unmanaged switches do not create additional broadcast domains; only a router (or a Layer-3 interface / VLAN) can separate them.

9.3 Improvements and Remaining Problem

Replacing hubs with switches produces the following benefits:

- Elimination of shared collision domains: each PC now has a dedicated full-duplex link, removing the possibility of collisions on the access segments.
- Higher effective throughput: switches can forward frames concurrently between different ports, utilising the full bandwidth of each link.
- Improved security and isolation: frames are forwarded only to the destination port (or flooded only when necessary), reducing unnecessary traffic exposure.
- Support for full-duplex operation: modern switches operate in full duplex by default, doubling potential bandwidth compared with half-duplex hub segments.

The problem that remains is the large size of the two broadcast domains. Broadcast traffic (ARP, DHCP, etc.) is still flooded to every device within each domain. In a production environment this would be mitigated by introducing VLANs on managed switches and/or additional Layer-3 boundaries, neither of which is present in the current alternative design.

10. Comparison of Original and Alternative Designs

Aspect	Original (Hubs)	Alternative (Switches)
Collision Domains	17	41
Broadcast Domains	2	2
Access-segment media access	Shared (CSMA/CD, half-duplex)	Dedicated (full-duplex)
Collision possibility on access links	Yes (within each hub)	No
Frame forwarding	Flood to all ports of hub	MAC-based switching
Effective bandwidth utilisation	Low (shared)	High (per-port)
Remaining limitation	Collisions + large broadcast domains	Large broadcast domains only

11. Conclusion

The original contractor design contains 17 collision domains and 2 broadcast domains. The three hubs each constitute a single shared collision domain, while every switch port and each router interface link forms its own collision domain. The router is the only device that separates the network into two broadcast domains.

Replacing the hubs with unmanaged switches increases the number of collision domains to 41 because each access port becomes an independent collision domain. Broadcast domains remain unchanged at two. The substitution eliminates collisions on the access segments, enables full-duplex operation and improves overall throughput and security. The residual limitation is the still-large broadcast domains, which would require VLANs or additional Layer-3 segmentation for further optimisation.

Standards-compliant equipment is essential for interoperability, security, cost-effectiveness and long-term maintainability of enterprise networks.